# Dr. James R. Beattie

updated: 11-September-2025

## Personal Information


| | |
|---|---|
| Nationality: | · Australia · New Zealand |
| Positions: | · Joint Princeton & CITA Postdoctoral Fellow in astrophysical plasmas |
| email: | · james.beattie@princeton.edu · jbeattie@cita.utoronto.edu |
| Online Profiles | · Google Scholar · ResearchGate · OrcID · Twitter |
| Interests: | · MHD / HD turbulence · structure of the interstellar medium · star-formation |
| | · high-performance computing · theoretical astrophysics · magnetic fields |
| | · cosmic ray propagation · plasma/fluid dynamics · shocks · turbulent dynamo |
| | · computer vision techniques · statistical modelling · interdisciplinary research |


## Education

2019-2024 **Doctor of Philosophy**, Australian National University, Australia
**Specialisation:** Computational / theoretical astrophysics, magnetohydrodynamics.
**Thesis**: The statistics of magnetised interstellar turbulence
**Advisor:** Christoph Federrath

2018-2019 **Honours (First Class)**, Australian National University, Australia
**Major**: Astrophysics
**Thesis**: Supersonic Turbulent Molecular Clouds: Filaments and Anisotropies
(with University Medal, Chancellor's Commendations, Bok Prize)

2013-2018 **Bachelor of Mathematics**, Queensland University of Technology, Australia
**Major**: Applied and Computational Mathematics

2013-2018 **Bachelor of Science**, Queensland University of Technology, Australia
**Major:** Physics

2008-2013 **Bachelor of Education**, Queensland University of Technology, Australia
**Major**: Biology & Computing

**Exchange and Summer Programmes**

| | |
|---|---|
| 2022-23 | Fulbright Exchange at the University of California, Santa Cruz, United States |
| 2017-18 | Cross Institutional Exchange at the University of Queensland, Australia |
| Winter 2017 | Summer Science Programme at The University of Cambridge, United Kingdom |
| Fall 2015 | Exchange Semester at Simon Fraser University, Canada |

## Scholarships, Awards & Grants

**Selected Scholarships & Fellowships**

| | |
|---|---|
| 2023 | CITA Fellow, CITA |
| 2023 | Research Associate, Princeton University |
| 2023 | Stanford Science Fellow, Stanford (declined) |
| 2022 | Fulbright PhD Fellowship |
| 2020 | Joan Duffield Research Supplementary Scholarship |
| 2020 | Deakin PhD Scholarship |
| 2020 | Dean's Merit (theoretical physics) HDR Supplementary Scholarship |
| 2019 | Bok Honours Astrophysics Scholarship |

**Selected Significant Awards**

| | |
|---|---|
| 2020 | Chancellor's Letter of Commendation: 7.0/7.0 Honours GPA |
| 2020 | ASA Bok Prize: Best Astronomy Honours Thesis in Australia |
| 2020 | Best Student Talk at ANITA, 2020 |
| 2019 | University Medal (top in graduating science cohort) |
| 2018 | Admission to the Dean's List of Students with Excellent Academic Performance |
| 2018 | Vice Chancellor's Performance Award |
| 2018 | Nominated for 2018 Vice-Chancellor's Awards for Excellence |
| 2016 | Vice Chancellor's Performance Award: Best Sessional Teaching in Science & Engineering Faculty |

| | |
|---|---|
| 2016 | Admission to the Dean's List of Students with Excellent Academic Performance |
| 2015 | Admission to the Dean's List of Students with Excellent Academic Performance |
| 2014 | Admission to the Dean's List of Students with Excellent Academic Performance |

**Computing grants awarded (1 core hour $\approx \$0.13$)**

| | | |
|---|---|---|
| 2024 | (PI) LRZ regular call: KHI dynamos | 3e6 core hours |
| 2024 | (PI) LRZ large scale call: The world's largest MHD simulation | 7e7 core hours |
| 2022 | (PI) LRZ large scale call extension | 1.5e7 core hours |
| 2021 | (PI) LRZ large scale call: The world's largest MHD simulation | 7e7 core hours |

## Professional Activities & Organisation Affiliations

**Professional Referee Activities**

| | |
|---|---|
| (1 article) | Open Journal of Astrophysics |
| (2 articles) | Astrophysical Journal |
| (4 articles) | Monthly Notices of the Astronomical Society |
| (1 articles) | Monthly Notices of the Astronomical Society Letters |
| (1 article) | Publications of the Astronomical Society of the Pacific |
| (first call 2024) | LRZ SuperMUC-NG large-scale compute project calls |

**Other Professional Activities**

| | |
|---|---|
| 2023-present | Coordinator for the Canadian Institute of Astrophysics Astro-Plasma Group |
| 2022 | MSATT program – connecting scientists with high school students |
| 2022 | Sustainability Committee, Member, RSAA |
| 2021 | Higher Degree Research Education Representative, RSAA |
| 2021 | Giving Committee, Member, RSAA |
| 2020 | President of the RSAA Student Seminar Committee |

## Supervisions & Mentoring

**Supervisions**

2025 **Student**: Jaden Cabral Cordeiro (co-supervised w. Bart Ripperda), Undergrad. Student
**Institute**: University of Toronto
**Project:** The transport of helicity in MHD turbulence

2025 **Student**: Daniel Hagyard (co-supervised w. Bart Ripperda), Undergrad. Student
**Institute**: University of Toronto
**Project:** Understanding density gradient statistics for scintillation and laboratory astrophysics

2024-25 **Student**: Isabelle Conner (co-supervised w. Enrico Ramirez-Ruiz), Undergrad. Student
**Institute**: University of California, Santa Cruz
**Project:** The universality of supernovae-driven turbulence in different galaxies.

2024 **Student**: Max Sokolova (co-supervised w. Bart Ripperda), Undergrad. Student
**Institute**: University of Toronto
**Project:** The MHD fluid and nonlinear vector Shrödinger correspondence

2024 **Student**:Juan Miguel L. Padilla (co-supervised w. Bart Ripperda), Undergrad. Student
**Institute**: University of Toronto
**Project:** Coherent vortices in MHD turbulence

2024 **Student**: Louis Burnaz (co-supervised w. Bart Ripperda), Undergrad. Student
**Institute**: École normale supérieure de Lyon
**Project:** Compression-triggered fast reconnection in relativistic, resistive MHD

2023-24 **Student**: Shashvat Varma (co-supervised w. Bart Ripperda), Undergrad. Student
**Institute**: University of Toronto
**Project:** The fast-in-time dynamics of the small-scale dynamo

2023 **Student**: Sam Lakerdas-Gayle (co-supervised w. Bart Ripperda), Undergrad. Student
**Institute**: University of Toronto
**Project:** The secret-life of over-dense regions in magnetised, turbulent clouds

2021 **Student**: Neco Kriel (co-supervised w. Christoph Federrath), Honours Student
**Institute**: Australian National University
**Project:** Fundamental scaling relations in the turbulent dynamo.

2021 **Student**: Matthew Sampson (co-supervised w. Mark Krumholz), Honours Student
**Institute**: Australian National University
**Project:** Cosmic ray transport in compressible ionised MHD turbulence.

**Mentorships**

2022 **Student**: Adrian Lehane
**Institute**: Telopea Park School / Narrabundah College (high school)
**Project:** Automated phase detection of Venus.

## Talks

**Invited (29 total)**

FEB. 2026 Astronomy Seminar, Michigan State

OCT. 2025 Scintillometry 2025, Montreal

SEP. 2025 MIST2025 - Cosmic turbulence and Magnetic fields: physics of baryonic matter

SEP. 2025 AAPS, Plasma Division, Japan

AUG. 2025 The chronology of high-$k$ magnetic modes from dynamo to stationary turbulence, Presented at: Frontiers of Relativistic Plasmas Physics in Astrophysics and Laboratory, KITP, Santa Barbara

JULY. 2025 Magnetized supersonic turbulence with unprecedented dynamical range, Presented at: ASTRONUM, University of Wisconsin, Madison

APRIL. 2025 No phenomenology is safe: surprises from extreme Reynolds number MHD turbulence simulations. Presented at: Caltech tea talk.

DEC. 2024 Supernova-driven turbulence in a multiphase ISM. Presented at: CITA blackboard talk.

NOV. 2024 Unravelling the workings of the compressible turbulent dynamo with $10{,}080^3$ simulations. Presented at: Princeton Plasma Physics Lab, Heliophysics Seminar.

SEP. 2024 The energy cascade II: supernova-driven turbulence. Goutelas, France.

SEP. 2024 The energy cascade I: the world's largest MHD turbulence box. Goutelas, France.

APR. 2024 Growing a magnetic field in a turbulent plasma. Presented at: CITA Blackboard Seminar.

APR. 2024 Supersonic magnetised turbulence at extreme Reynolds numbers. Presented at: CITA Theory Seminar

MAR. 2024 Unravelling the fast growth stages of the turbulent dynamo. Presented at: University Maryland Computational and Theory Seminar

FEB. 2024 Numerical resistivity and viscosity. KITP Talk.

FEB. 2024 A large compressible MHD turbulence simulation with no net magnetic flux. KITP Talk.

NOV. 2023 The most fascinating part of interstellar turbulence: the energy cascade Presented at: TASTY Seminar Series, University of Toronto.

MAY 2023 The World's Largest Compressible MHD Turbulence Simulation on SuperMUC-NG Presented at: SuperMUC-NG Status and Results Workshop.

SEP. 2022 KIPAC Tea talk: Peta-scale magnetised interstellar medium turbulence simulations. Presented at: SLAC / Stanford University.

SEP. 2022 Magnetised interstellar medium turbulence: dynamics & energetics. Presented at: Susan Clark's research group, Stanford.

SEP. 2022 Astro-coffee: Streaming cosmic rays ion Alfvén velocity statistics. Presented at: Institute for Advanced Study.

SEP. 2022 Bachall lunch discussion: peta-scale simulations & turbulent dynamics. Presented at: Institute for Advanced Study.

APR. 2022 Streaming cosmic rays ion Alfvén velocity statistics. Presented at: Siang Peng Oh's research group, UC Santa Barbara.

NOV. 2021 Ubiquitous magnetic field fluctuations driven by large-scale supersonic turbulence. Presented at: Star formation and ISM Physics Seminar, Princeton.

JAN. 2021 Ubiquitous magnetic field fluctuations driven by large-scale supersonic turbulence. Presented at: Research School of Astronomy and Astrophysics seminar, ANU.

JUL. 2020 The Anisotropic Density Variance for Highly-Magnetised Molecular Clouds. Presented at: Astronomical Society of Australia Bok Prize talk.

JUN. 2020 Turbulence at the parsec scale of the Universe. Presented at: Research highlight talk at RSAA full school meeting.

AUG. 2018 The Fractal Geometry of the Supersonic Turbulence in the Interstellar Medium. Presented at: QUT research highlights.

MAY 2018 The Fractal Geometry of Turbulence. Presented at: QUT Physics Society Meeting.

**Colloquium (3 total)**

Aug. 2020 The Anisotropic Density Variance for Highly-Magnetised Molecular Clouds.
Presented at: University of Macquarie Colloquium.

Nov. 2017 The University of Cambridge and Quantum Mechanics.
Presented at: School of Chemistry, Physics and Engineering Colloquium, QUT.

Nov. 2017 Mathematical Aspects of Mechanics.
Presented at: School of Mathematical Sciences Colloquium, QUT.

**Contributed (17 total)**

Sep. 2024 From Fast Growth to the Saturation of the ICM dynamo
Presented at: Galaxy Clusters and Radio Relics II 2024 Workshop, Harvard, Boston

May. 2024 Interstellar medium turbulence and turbulence-regulated star formation theory
Presented at: Globular Clusters and Their Tidal Tails: From the Milky Way to the Local Group, Toronto, Canada

May. 2024 The Supersonic Turbulent Dynamo
Presented at: HEDLA 2024 Workshop, Tallahassee, Florida

Feb. 2022 Petascale magnetised interstellar medium turbulence simulations
Presented at: ANITA 2022 Workshop.

Dec. 2021 Understanding the nature of magnetic field fluctuations driven by large-scale supersonic turbulence.
Presented at: Australian Institute of Physics Congress, QUT.

Oct. 2021 Understanding the nature of magnetic field fluctuations driven by large-scale supersonic turbulence.
Presented at: Royal Astronomical Society: Galactic magnetic fields meeting.

Feb. 2021 Steps towards anisotropic star formation theory: A multi-shock model for the density variance of anisotropic MHD turbulence.
Presented at: ANITA 2021 Workshop.

Dec. 2020 Multi-shock model for the density variance of anisotropic, highly-magnetised ISM turbulence.
Presented at: The Magnetic Field Awakens: A new era of star formation.

Nov. 2020 Recent progress on anisotropic, magnetised, supersonic turbulence.
Presented at: Mount Stromlo Student Seminars, 2020.

Sep. 2020 Is the Starry Night Turbulent?
Presented at: RSAA Feast of Facts.

Feb. 2020 Density, velocity and magnetic structures and correlations in sub-Alfvénic, supersonic turbulence.
Accepted* for contributed talk: Magnetic Fields in the Universe 7, Vietnam.

Feb. 2020 Anisotropy in the column density of highly-magnetised supersonic turbulence.
Presented at: ANITA 2020 Workshop, UNSW, Canberra.

Dec. 2019 Anisotropic structures in highly-magnetised, observed turbulent clouds.
Presented at: Universality: Turbulence across vast scales, Flatiron Inst., New York

Nov. 2019 Reconstructing the 3D Density PDF from the 2D Column Density.
Presented at: Cosmic turbulence and magnetic fields : physics of baryonic matter across time and scales in Cargese, France, 2019.

Dec. 2018 Mach number - fractal dimension relation for turbulent, molecular clouds.
Poster presented at: AIP Congress 2018, Perth, Australia.

Jan. 2018 The Fractal Geometry of the World's Largest Turbulence Simulation.
Presented at: Research School of Astronomy and Astrophysics, ANU.

Jan. 2017 The Analysis of Novel Magnetic Field Configurations in the H-1 NF Stellarator.
Presented at: Research School of Physics and Engineering, ANU.

* did not attend due to COVID19

**Public Outreach Talks (10 total)**

Aug. 2021 Building the Universe, Brick-by-brick. Presented at: Young Stars, ANU, Canberra.

May. 2021 Understanding The Big Bang. Presented at: Young Stars, ANU, Canberra.

Mar. 2021 The Secret Life of Cells. Presented at: Young Stars, ANU, Canberra.

Jan. 2021 Mission to Mars. Presented at: Young Stars, ANU, Canberra.

Jan. 2021 The Jiggling Universe. Presented at: SciScouts Space Squad, Canberra.

Nov. 2020 The Jiggling Universe. Presented at: Campbell Primary School STEM day, Canberra.

Oct. 2020 Thinking Like An Atom. Presented at: Young Stars, Canberra.

Sep. 2020 Simulating the Universe. Presented at: SciScouts Space Squad, Canberra.

Mar. 2020 Modelling Pandemics. Presented at: Young Stars, Canberra.

Feb. 2020 How do scientists test their ideas? Presented at: Young Stars, Canberra.

## Teaching (23 total contributions)

### Guest Lectures

Oct. 2022 ASTR8002 (ANU): Guest lecture on MHD turbulence theory for a graduate level gas dynamics class.

Oct. 2020 ASTR8002 (ANU): Guest lecture on linear MHD waves for a graduate level gas dynamics class.

### TA experience (Click on the "Semester" to see teacher evaluation reports)

2021 **Australian National University**, Canberra, Australia
ASTR2013: Foundations of Astrophysics — Semester Two

2018 **Queensland University of Technology**, Brisbane, Australia
PVB101: Physics of the Large — Semester Two
MXB105: Calculus of One and Two Variables (wrote all assessment) — Semester Two
MXB161: Computational Explorations — Semester One
SEB113: Quantitative Methods in Science — Semester One & Two
SEB104: Grand Challenges in Science — Semester One
SEB115: Experimental Science — Semester One

2017 **Queensland University of Technology**, Brisbane, Australia
MXB105: Calculus of One and Two Variables — Semester Two
PVB101: Physics of the Large (Lab Demonstrator) — Semester Two
BVB204: Ecology — Semester Two
SEB113: Quantitative Methods in Science — Semester One & Two
SEB104: Grand Challenges in Science — Semester One
SEB115: Experimental Science (Lab Demonstrator) — Semester One
MXB161: Computational Explorations — Semester One

2016 **Queensland University of Technology**, Brisbane, Australia
PVB101: Physics of the Large (Lab Demonstrator) — Semester Two
BVB202: Plant Biology (Lab Demonstrator) — Semester Two
BVB224: Plant Diversity (Lab Demonstrator) — Semester Two
SEB113: Quantitative Methods in Science — Semester One & Two
SEB104: Grand Challenges in Science — Semester One
SEB115: Experimental Science (Lab Demonstrator) — Semester One

2015 **Queensland University of Technology**, Brisbane, Australia
SEB113: Quantitative Methods in Science — Semester One

## Publications

- **First (and joint first) author:** 18 publications (14 refereed)
- **Total:** 37 publications
- **Citations:** 850 (11-Sep-2025)
- **h index:** 16 (11-Sep-2025)

### First Author (and joint first) Refereed (14 total)

**Beattie, J. R.**, Federrath, C., Kriel, N., Hew, J. K. J., & Bhattacharjee, A. (2025). Taking control of compressible modes: bulk viscosity and the turbulent dynamo. *MNRAS, accepted*, Article arXiv:2312.03984, arXiv:2312.03984. https://doi.org/10.48550/arXiv.2312.03984

**Beattie, J. R.**, Federrath, C., Klessen, R. S., Cielo, S., & Bhattacharjee, A. (2025). The spectrum of magnetized turbulence in the interstellar medium. *Nature Astronomy*. https://doi.org/10.1038/s41550-025-02551-5

Bandyopadhyay, R., **Beattie, J. R.**, & Bhattacharjee, A. (2025). Density fluctuation–mach number scaling in compressible, high plasma beta turbulence: In situ space observations and high-reynolds number simulations. *The Astrophysical Journal Letters*, *982*(2), L45. https://doi.org/10.3847/2041-8213/adbe3b

**Beattie, J. R.**, Noer Kolborg, A., Ramirez-Ruiz, E., & Federrath, C. (2025). So long Kolmogorov: the forward and backward turbulence cascades in a supernovae-driven, multiphase interstellar medium. *accepted in ApJ*, Article arXiv:2501.09855, arXiv:2501.09855. https://doi.org/10.48550/arXiv.2501.09855

**Beattie, J. R.**, Federrath, C., Kriel, N., Mocz, P., & Seta, A. (2023). Growth or decay – I: universality of the turbulent dynamo saturation. *arXiv e-prints*, arXiv:2209.10749.

Birch, M., **Beattie, J. R.**, Bennet, F., Rattenbury, N., Copeland, M., Travouillon, T., Ferguson, K., Cater, J., & Sayat, M. (2023). Availability, outage, and capacity of spatially correlated, australasian free-space optical networks. *J. Opt. Commun. Netw.*, *15*(7), 415–430. https://doi.org/10.1364/JOCN.480805

**Beattie, J. R.**, Krumholz, M. R., Federrath, C., Sampson, M. L., & Crocker, R. M. (2022). Ion alfvén velocity fluctuations and implications for the diffusion of streaming cosmic rays. *Frontiers in Astronomy and Space Sciences*, *9*. https://doi.org/10.3389/fspas.2022.900900

**Beattie, J. R.**, Mocz, P., Federrath, C., & Klessen, R. S. (2022). The density distribution and physical origins of intermittency in supersonic, highly magnetised turbulence with diverse modes of driving. *MNRAS*. https://doi.org/10.1093/mnras/stac3005

**Beattie, J. R.**, Krumholz, M. R., Skalidis, R., Federrath, C., Seta, A., Crocker, R. M., Mocz, P., & Kriel, N. (2022). Energy balance and Alfvén Mach numbers in compressible magnetohydrodynamic turbulence with a large-scale magnetic field. *MNRAS*. https://doi.org/10.1093/mnras/stac2099

**Beattie, J. R.**, Mocz, P., Federrath, C., & Klessen, R. S. (2021). A multishock model for the density variance of anisotropic, highly magnetized, supersonic turbulence. *MNRAS*, *504*(3), 4354–4368. https://doi.org/10.1093/mnras/stab1037

**Beattie, J. R.**, Federrath, C., & Seta, A. (2020). Magnetic field fluctuations in anisotropic, supersonic turbulence. *MNRAS*, *498*(2), 1593–1608. https://doi.org/10.1093/mnras/staa2257

**Beattie, J. R.**, & Federrath, C. (2020). Filaments and striations: anisotropies in observed, supersonic, highly magnetized turbulent clouds. *MNRAS*, *492*(1), 668–685. https://doi.org/10.1093/mnras/stz3377

**Beattie, J. R.**, Federrath, C., Klessen, R. S., & Schneider, N. (2019). The relation between the turbulent Mach number and observed fractal dimensions of turbulent clouds. *MNRAS*, *488*(2), 2493–2502. https://doi.org/10.1093/mnras/stz1853

**Beattie, J. R.**, Federrath, C., & Klessen, R. S. (2019). The relation between the true and observed fractal dimensions of turbulent clouds. *MNRAS*, *487*(2), 2070–2081. https://doi.org/10.1093/mnras/stz1416

**Second Author or Major Contributions Refereed (9 total)**

Kriel, N., **Beattie, J. R.**, Federrath, C., Krumholz, M. R., & Hew, J. K. J. (2025). Fundamental MHD scales - II: the kinematic phase of the supersonic small-scale dynamo. *MNRAS*. https://doi.org/10.1093/mnras/staf188

Sampson, M. L., **Beattie, J. R.**, Krumholz, M. R., Crocker, R. M., Federrath, C., & Seta, A. (2023). Turbulent diffusion of streaming cosmic rays in compressible, partially ionized plasma. *MNRAS*, *519*(1), 1503–1525. https://doi.org/10.1093/mnras/stac3207.

**Ran all MHD turbulence models, provided analytical Green's function solutions to the diffusion problems, helped develop the theory and fitting for fractional diffusion transport and contributed to writing the manuscript.**

Kriel, N., **Beattie, J. R.**, Seta, A., & Federrath, C. (2022). Fundamental scales in the kinematic phase of the turbulent dynamo. *MNRAS*. https://doi.org/10.1093/mnras/stac969.

**Developed the spectral fitting methodology, spectral models, taught Kriel how to use the** FLASH **code throughout the project and contributed to writing the manuscript.**

Risch, A. C., Page-Dumroese, D. S., Schweiger, A. K., **Beattie, J. R.**, Curran, M. P., Finér, L., Liu, Y., Schütz, M., Terry, T. A., Wang, W., & Jurgensen, M. F. (2022). Controls of initial wood decomposition on and in forest soils using standard material. *Frontiers in Forests and Global Change*, *5*, 829810. https://doi.org/10.3389/ffgc.2022.829810.

**Constructed the principle data set, developed and ran parallelised hierarchical Bayesian mixed effects models and model selection methods.**

Skalidis, R., Sternberg, J., **Beattie, J. R.**, Pavlidou, V., & Tassis, K. (2021). Why take the square root? An assessment of interstellar magnetic field strength estimation methods. *A&A*, *656*, Article A118, A118. https://doi.org/10.1051/0004-6361/202142045.

**Ran all MHD turbulence simulations and contributed to the theoretical development of the coupling term energy model and drafting the manuscript.**

Federrath, C., Klessen, R. S., Iapichino, L., & **Beattie, J. R.** (2021). The sonic scale of interstellar turbulence. *Nature Astronomy*, *5*, 365–371. https://doi.org/10.1038/s41550-020-01282-z.

**Measured the sonic scale position from the second order structure functions and contributed to writing the manuscript.**

Thomas, M. L., Baker, L., **Beattie, J. R.**, & Baker, A. M. (2020). Determining the efficacy of camera traps, live capture traps, and detection dogs for locating cryptic small mammal species. *Ecology and Evolution*, *10*(2), 1054–1068. https://doi.org/10.1002/ece3.5972.

**Developed and applied the occupancy analysis models used to compare between the different detection methods and contributed to writing the manuscript.**

McCool, C., **Beattie**, **J. R.**, Milford, M., Bakker J. D., J. L., Moore, & Firn, J. (2018). Automating analysis of vegetation with computer vision: Cover estimates and classification. *Ecology and Evolution*, *8*(12), 6005–6015. https://doi.org/10.1002/ece3.4135.

**Developed and applied the statistical model for comparing between the different computer vision techniques and contributed to writing the manuscript.**

McCool, C., **Beattie**, **J. R.**, Firn, J., Lehnert, C., Kulk, J., Bawden, O., Russell, R., & Perez, T. (2018). Efficacy of mechanical weeding tools: A study into alternative weed management strategies enabled by robotics. *IEEE Robotics and Automation Letters*, *3*(2), 1184–1190. https://doi.org/10.1109/LRA.2018.2794619.

**Developed and applied the survival analysis models used to compare between the different automated weeding strategies and contributed to writing the manuscript.**

### Multi-author Refereed (4 total)

Schneider, N., Ossenkopf-Okada, V., Clarke, S., Klessen, R. S., Kabanovic, S., Veltchev, T., Bontemps, S., Dib, S., Csengeri, T., Federrath, C., Di Francesco, J., Motte, F., André, Ph., Arzoumanian, D., Beattie, J. R., Bonne, L., Didelon, P., Elia, D., Könyves, V., ... Ward-Thompson, D. (2022). Understanding star formation in molecular clouds - iv. column density pdfs from quiescent to massive molecular clouds. *A&A*, *666*, A165. https://doi.org/10.1051/0004-6361/202039610

Seligman, D. Z., Rogers, L. A., Feinstein, A. D., Krumholz, M. R., **Beattie**, **J. R.**, Federrath, C., Adams, F. C., Fatuzzo, M., & Günther, M. N. (2022). Theoretical and Observational Evidence for Coriolis Effects in Coronal Magnetic Fields via Direct Current Driven Flaring Events. *ApJ*, *929*(1), Article 54, 54. https://doi.org/10.3847/1538-4357/ac5b69

Sharda, P., Menon, S. H., Federrath, C., Krumholz, M. R., **Beattie**, **J. R.**, Jameson, K. E., Tokuda, K., Burkhart, B., Crocker, R. M., Law, C. J., Seta, A., Gaetz, T. J., Pingel, N. M., Seitenzahl, I. R., Sano, H., & Fukui, Y. (2022). First extragalactic measurement of the turbulence driving parameter: ALMA observations of the star-forming region N159E in the Large Magellanic Cloud. *MNRAS*, *509*(2), 2180–2193. https://doi.org/10.1093/mnras/stab3048

Milford, M., Firn, J., **Beattie**, **J.**, Jacobson, A., Pepperell, E., Mason, E., Kimlin, M., & Dunbabin, M. (2014). Automated sensory data alignment for environmental and epidermal change monitoring. *Australasian Conference on Robotics and Automation 2014*, 1–10. https://eprints.qut.edu.au/81684/

### Preprints Undergoing Review or Other (10 total)

**Beattie**, **J. R.**, & Bhattacharjee, A. (2025). Scale-dependent alignment and the depletion of nonlinearities in compressible magnetohydrodynamic turbulence. *Submitted to PRL*.

Connor, I., **Beattie**, **J. R.**, Kolborg, A. N., & Ramirez-Ruiz, E. (2025). Cascading from the winds to the disk: The universality of supernovae-driven turbulence in different galactic interstellar mediums. https://arxiv.org/abs/2509.01653

**Beattie**, **J. R.** (2025). Supernovae drive large-scale, incompressible turbulence through small-scale instabilities. *arXiv e-prints*, Article arXiv:2509.07354, arXiv:2509.07354. https://doi.org/10.48550/arXiv.2509.07354

Kriel, N., **Beattie**, **J. R.**, Krumholz, M. R., Schober, J., & Armstrong, P. J. (2025). The growth of magnetic energy during the nonlinear phase of the subsonic and supersonic small-scale dynamo. *Submitted to PRL*, Article arXiv:2509.09949, arXiv:2509.09949.

Hew, J. K. J., Hosking, D. N., Federrath, C., **Beattie**, **J. R.**, & Kriel, N. (2025). Conservation of magnetic-helicity fluctuations due to spatial decorrelation of fluxes in decaying mhd turbulence. *Submitted to Journal of Plasma Physics*.

Sampson, M. L., **Beattie**, **J. R.**, Teyssier, R., Kempski, P., Moseley, E. R., Commerçon, B., Dubois, Y., & Rosdahl, J. (2025). Cosmic ray and plasma coupling for isothermal supersonic turbulence in the magnetized interstellar medium. *arXiv e-prints*, Article arXiv:2506.03768, arXiv:2506.03768. https://doi.org/10.48550/arXiv.2506.03768

Grehan, M. P., Ghosal, T., **Beattie**, **J. R.**, Ripperda, B., Porth, O., & Bacchini, F. (2025). Comparison of magnetic diffusion and reconnection in ideal and resistive relativistic magnetohydrodynamics, ideal magnetodynamics and resistive force-free electrodynamics. *arXiv e-prints*, Article arXiv:2503.20013, arXiv:2503.20013. https://doi.org/10.48550/arXiv.2503.20013

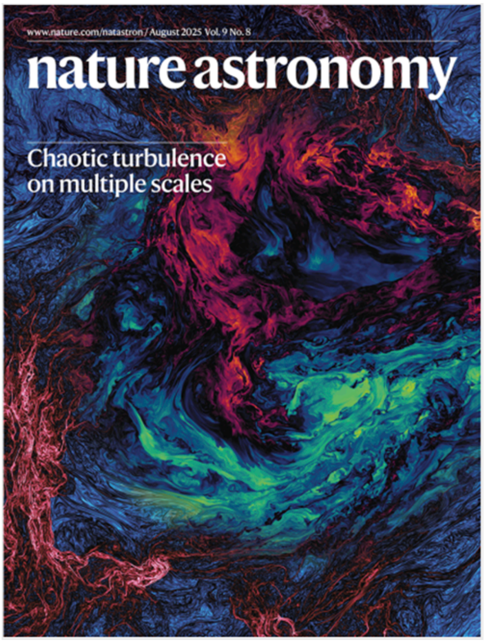
Bastian, P., Kranzlmüller, D., Brüchle, H., & Mathias, G. (Eds.). (2024). *High performance computing in science and engineering – garching/munich 2024*. Leibniz-Rechenzentrum der Bayerischen Akademie der Wissenschaften. https://www.lrz.de/hpcbooks/

**Beattie, J. R.**, Federrath, C., Klessen, R. S., Cielo, S., & Bhattacharjee, A. (2024). Magnetized compressible turbulence with a fluctuation dynamo and Reynolds numbers over a million. *arXiv e-prints*, Article arXiv:2405.16626, arXiv:2405.16626. https://doi.org/10.48550/arXiv.2405.16626

**Beattie, J. R.**, & Kriel, N. (2019). Is The Starry Night Turbulent? *arXiv e-prints*, arXiv:1902.03381.

## Media (25 total)

**2025** Nature Astronomy cover for August 2025 issue: Chaotic turbulence on multiple scales.




**2025** Space Turbulence, *New Zealand National Geographic*

**2025** Eddies, Flows and van Gogh: Probing the Interstellar Medium, *Centauri Dreams*

**2025** Scientists precisely simulate turbulence in the Galaxy – it doesn't behave like they thought, *Eurekalert*

**2025** Scientists Map Hidden Patterns of Magnetized Turbulence in Interstellar Space, *University of Heidelberg*

**2025** Hidden Patterns of Cosmic Turbulence unveiled, *University of Heidelberg*

**2025** Astrophysicists are exploring our galaxy's magnetic turbulence in unprecedented detail using a new computer model, *University of Toronto*

**2025** Astrophysicists explore our galaxy's magnetic turbulence in unprecedented detail using a new computer model, *phys.org*

**2025** Commentary for Scientists make 'rare advance' in tackling the oldest unsolved problem in physics, *CNN*

**2024** The Hidden Science Swirling in The Starry Night, *Washington Post*

**2024** CITA Fellow James Beattie was awarded close to 100-million-core-hours to explore the nature of turbulence and dynamos, *CITA*

**2024** Van Gogh's depiction of the starry night closely resembles the turbulence found between the stars, *CITA*

**2024** Expert quoted in "Turbulent skies of Vincent Van Gogh's 'The Starry Night'", *CNN*

**2024** Quoted in "The Unexpected Poetry of PHD Thesis Acknowledgements", *Australian National University*

**2024** Plasma: beautiful chaos, *Leibniz Supercomputing Centre July Newsletter*

**2024** The Beauty of Chaos, *New Scientist Magazine (in print and online)*

**2024** The world's largest magnetohydrodynamic turbulence simulation, *Forschung Magazine*

**2024** 2024 Research Highlight Calendar (March visualisation), *German Research Foundation (DFG)*

**2022** Unravelling magnetised turbulence in galaxies, *Lunations, Research Bytes*

**2022** The Magic And Mystery Of Turbulence, *IFL Science*

**2021** Extreme efficiency astrophysical turbulence simulations, *National Computing Infrastructure, Australia*

**2021** Coffee, planes and magnetism. *Space Australia, TikTok*

**2021** Unravelling the turbulent, magnetised dynamics of the interstellar medium. *Space Australia*

**2021** Turbulence in the heavens, *Nature Astronomy, News & Views*

**2021** Researchers Use LRZ HPC Resources to Perform Largest-Ever Supersonic Turbulence Simulation, *Gauss Centre for Supercomputing*

**2021** The Role of Turbulence in the Birth of Stars, *University Heidelberg*

**2021** Star-making motion, *COSMOS magazine*

**2021** Study helps unlock secrets of star formation, *ANU Media*

**2021** Stellar Simulation Reveals The Turbulent Nature of Star Birth, *Space Australia*

**2021** The Need for (Sound) Speed, *Astrobite research highlight*

**2019** Modelling Star Formation with a Supercomputer: Computational Astrophysics Research, *National Computing Infrastructure Australia*

**2019** Feature article on turbulence depicted in Van Gogh's Starry Night in the Art's and Culture section of the American Physical Societies Magazine.

**2018** QUT advertising photoshoot for the BSc and BMath degree on QUT's blackboard website.

**2018** QUT media exposure, and photoshoot for our publication, Automating Analysis of Vegetation with Computer Vision: cover estimates.